# Small Signal Schottky Diode



## FEATURES

- For general purpose applications
- These diodes feature very low turn-on voltage and fast switching
- These devices are protected by a PN junction guard ring against excessive voltage, such as electrostatic discharges
- These diodes are also available in the DO-35 (DO-204AH) case with type designations BAT42 to BAT43 and in the SOD-123 case with type designations BAT42W-V to BAT43W-V
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT

## LINKS TO ADDITIONAL RESOURCES

3D Models | Marking | Parametric Search | Order Samples

## MECHANICAL DATA

**Case:** MiniMELF (SOD-80)

**Weight:** approx. 31 mg

**Cathode band color:** black

**Packaging codes/options:**

GS18/10K per 13" reel (8 mm tape), 10K/box

GS08/2.5K per 7" reel (8 mm tape), 12.5K/box

**PARTS TABLE**

| PART | ORDERING CODE | CIRCUIT CONFIGURATION | REMARKS |
|---|---|---|---|
| LL42 | LL42-GS18 or LL42-GS08 | Single | Tape and reel |
| LL43 | LL43-GS18 or LL43-GS08 | Single | Tape and reel |

**ABSOLUTE MAXIMUM RATINGS** ($T_{amb}$ = 25 °C, unless otherwise specified)

| PARAMETER | TEST CONDITION | SYMBOL | VALUE | UNIT |
|---|---|---|---|---|
| Repetitive peak reverse voltage | | $V_{RRM}$ | 30 | V |
| Forward continuous current (1) | | $I_F$ | 200 | mA |
| Repetitive peak forward current (1) | $t_p$ < 1 s, δ < 0.5 | $I_{FRM}$ | 500 | mA |
| Surge forward current (1) | $t_p$ = 10 ms | $I_{FSM}$ | 4 | A |
| Power dissipation (1) | $T_{amb}$ = 65 °C | $P_{tot}$ | 200 | mW |

**Note**

(1) Valid provided that electrodes are kept at ambient temperature

**THERMAL CHARACTERISTICS** ($T_{amb}$ = 25 °C, unless otherwise specified)

| PARAMETER | TEST CONDITION | SYMBOL | VALUE | UNIT |
|---|---|---|---|---|
| Thermal resistance junction to ambient air (1) | | $R_{thJA}$ | 300 | K/W |
| Junction temperature | | $T_j$ | 125 | °C |
| Ambient operating temperature range | | $T_{amb}$ | -55 to +125 | °C |
| Storage temperature range | | $T_{stg}$ | -65 to +150 | °C |

**Note**

(1) Valid provided that electrodes are kept at ambient temperature



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| **ELECTRICAL CHARACTERISTICS** ($T_{amb}$ = 25 °C, unless otherwise specified) | | | | | | | |
|---|---|---|---|---|---|---|---|
| **PARAMETER** | **TEST CONDITION** | **PART** | **SYMBOL** | **MIN.** | **TYP.** | **MAX.** | **UNIT** |
| Reverse breakdown voltage | $I_R$ = 100 µA (pulsed) | | $V_{(BR)}$ | 30 | | | V |
| Leakage current (1) | $V_R$ = 25 V | | $I_R$ | | | 0.5 | µA |
| | $V_R$ = 25 V, $T_j$ = 100 °C | | $I_R$ | | | 100 | µA |
| Forward voltage (1) | $I_F$ = 200 mA | | $V_F$ | | | 1000 | mV |
| | $I_F$ = 10 mA | LL42 | $V_F$ | | | 400 | mV |
| | $I_F$ = 50 mA | LL42 | $V_F$ | | | 650 | mV |
| | $I_F$ = 2 mA | LL43 | $V_F$ | 260 | | 330 | mV |
| | $I_F$ = 15 mA | LL43 | $V_F$ | | | 450 | mV |
| Diode capacitance | $V_R$ = 1 V, f = 1 MHz | | $C_D$ | | 7 | | pF |
| Reverse recovery time | $I_F$ = 10 mA, $I_R$ = 10 mA, $i_R$ = 1 mA, $R_L$ = 100 Ω | | $t_{rr}$ | | | 5 | ns |
| Rectification efficieny | $R_L$ = 15 kΩ, $C_L$ = 300 pF, f = 45 MHz, $V_{RF}$ = 2 V | | $\eta_v$ | 80 | | | % |

**Note**

(1) Pulse test $t_p$ < 300 µs, $t_p/T$ < 0.02

## TYPICAL CHARACTERISTICS ($T_{amb}$ = 25 °C, unless otherwise specified)

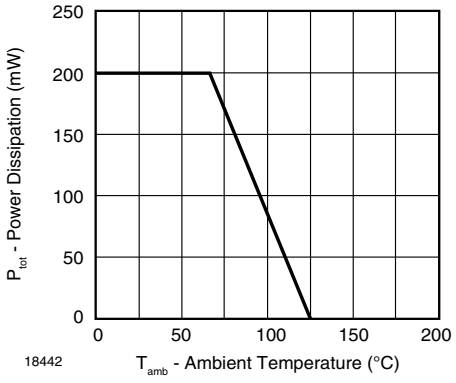



Fig. 1 - Admissible Power Dissipation vs. Ambient Temperature

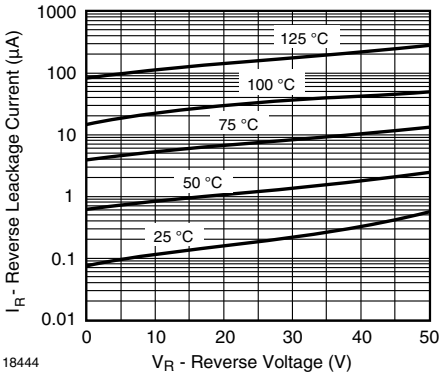



Fig. 3 - Typical Reverse Characteristics

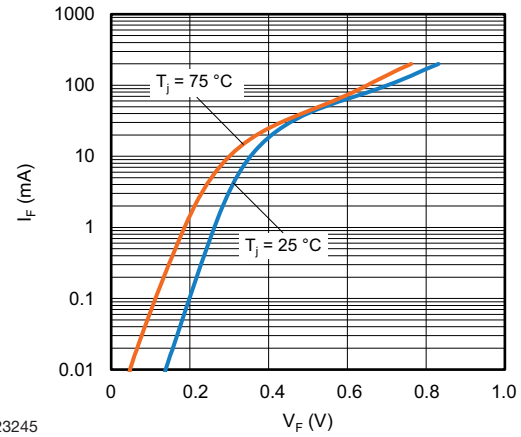



Fig. 2 - Typical Forward Characteristics

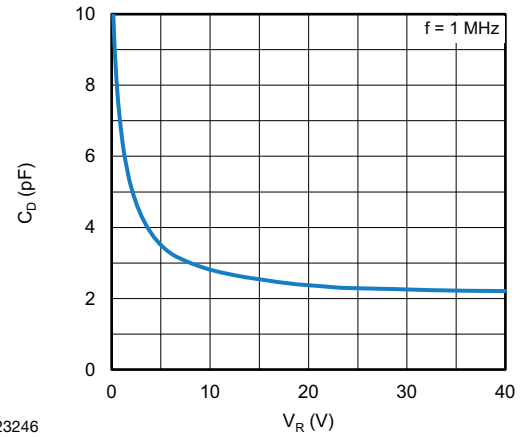


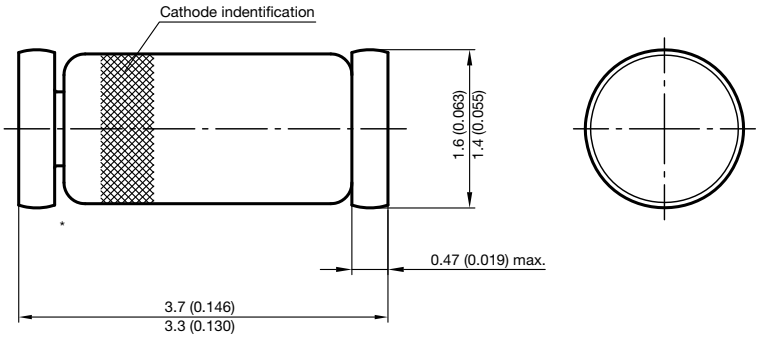

Fig. 4 - Typical Capacitance vs. Reverse Voltage

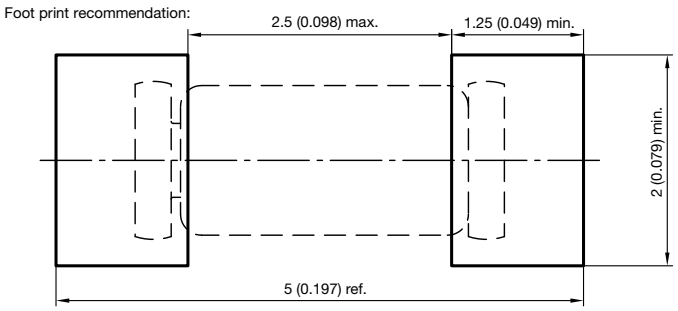


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**PACKAGE DIMENSIONS** in millimeters (inches): **MiniMELF (SOD-80)**




Document no.:6.560-5005.01-4
Rev. 8 - Date: 07.June.2006
96 12070


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